

Milad Azami

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Detail-oriented PhD in Materials Science and Engineering with over 5+ years of experience in Li-ion battery development and 3+ years of experience as a process engineer in high-volume wafer manufacturing. Extensive expertise in materials characterization, electrochemical testing, and failure analysis combined with proven ability to monitor and analyze real-time data in manufacturing environments. Demonstrated success in leading cross-functional teams, optimizing complex processes, and developing innovative solutions from laboratory research through industrial implementation.

Technical Skills

- Materials Testing and Characterization: XRD, SEM, TEM, TGA, Dilatometry, FTIR
- Li ion Batteries Research and Development: Slurry Formulation, Coating, Battery Cell Manufacturing
- Electrochemical Testing: CV, EIS, Cycle Life, Rate Capability Testing
- Modeling and Simulation: COMSOL Multiphysics, Python
- Data Analysis and Modeling: SPC, Design of Experiment (DOE), Root Cause Analysis, Statistical Analysis, Gen AI
- Project Management: Cross-Functional Team Collaboration, Vendor Coordination, Documentation and SOPs
- Manufacturing Processes: High-volume wafer manufacturing, plating processes, quality control, defect analysis
- Process Engineering: Real-time Data Monitoring, Process Optimization, Tool Qualification, Error Recovery Procedures
- Problem Solving: Green Belt Lean 6 Sigma, 8D

Experience

Plating Module Process Engineer / Intel, OR

(Mar 2022 – Sep 2024)

Real-Time Process Monitoring and Tool Health Optimization

Monitor CEV PCSA final height data regularly to identify potential issues before they flagged, ensuring healthy tools and minimizing downtime. Lead root cause analysis on flagged tools using data from one-click reviews and assign appropriate team members to resolve issues efficiently. Collaborate with cross-functional teams to develop and maintain real-time monitoring criteria (E3) for plating modules, enhancing process control and performance. Develop and implement standard operating procedures for tool and process recovery, including wafer recovery using robots and teaching protocols. (Real-Time Data Analysis, Root Cause Analysis, Process Control, Cross-Functional Collaboration, SOP Development)

Plating Process Parameter Optimization and Defect Reduction

Led the evaluation of multiple power supply units and plating cells, identifying non-zero stray currents that affected Cu2 plating and developed corrective actions to minimize defects. Optimized plating parameters by testing various soak times and plating conditions, which led to the successful implementation of a reduced soak time process with minimal stray current impact. Collaborated with cross-functional teams to optimize plating tool performance, including the creation of RFCs for plating duet clamshell error recovery and ensuring alignment with RFC specifications. Managed RFC and BKM procedure updates for issues like back-end robot time-out errors and bath concentration fails, improving process reliability and tool recovery. Authored detailed white papers documenting process changes, pilot results, and technical justifications, contributing to the effective communication and implementation of revised plating procedures. Successfully coordinated vendor qualifications and chemistry management tool setup like LAU, ensuring timely completion of troubleshooting and qualification processes. (Process Optimization, Defect Analysis, Technical Documentation, Vendor Coordination, Equipment Qualification)

Battery Scientist / Moses Lake Industries, OR

(Dec 2020 – Mar 2022)

MIEC Binder Screening and Manufacturing Process Optimization

Conducted comprehensive screening and assessment of MIEC (Mixed Ionic Electronic Conductor) binders to enhance battery electrode performance and manufacturing efficiency. Optimized manufacturing methods through systematic evaluation of processing parameters and material interactions. Identified and established metrology protocols needed for proficient diagnoses of electrode quality and performance characteristics. Developed standard operating procedures for electrode fabrication processes and maintained advanced electrochemical test setups to ensure reliable results

Cross-Functional Team Leadership and Experimental Protocol Development

Established comprehensive experimental plans to assess electrode performance through systematic testing protocols and data analysis frameworks. Led multiple teams across different functional areas while establishing constructive collaboration with both external vendors and internal research groups. Acquired, operated, and maintained sophisticated electrode fabrication equipment and electrochemical test setups, ensuring optimal performance and reliability. Developed and implemented testing protocols that became standard practice for electrode performance evaluation and quality assessment. (Team Leadership, Experimental Design, Equipment Management, Protocol Development, Vendor Collaboration)

Postdoctoral Fellow / Clemson University, Clemson, SC

(May 2020 – Dec 2020)

Conducted advanced research on battery failure analysis and materials development. Investigated failure mode, mechanism and analysis (FMMEA) of NMC811 half-cell and full-cell systems through various analytical methods including pOCV, electrochemical cycling, EIS, and IC-DV techniques. Investigated and illustrated the effect of electrolyte additives on degradation of NMC/graphite full cells.

Research Assistant, PhD / Clemson University, Clemson, SC

(Jan 2015 – Apr 2020)

Novel Processing Technique Development for High Performance Li-ion Battery Electrodes

Developed a novel, practical processing route that combines tape casting with freeze casting to engineer ultra-thick lithium-ion battery electrodes with ordered macro-porosity channels, enhancing electrolyte transport and enabling high power/energy densities. Designed and executed experiments to extract key electrochemical parameters—electrode and separator tortuosity, open-circuit voltage (OCV), Li-ion diffusivity as a function of state of charge (SOC), and reaction kinetics—and translated the results into optimized electrode formulations and builds (coin and Swagelok cells). Evaluated performance using cyclic voltammetry (CV), electrochemical impedance spectroscopy (EIS), cycle-life, and rate-capability testing, complemented by materials characterization and robust process documentation to support repeatable manufacturing and scale-up.

MoO₃ Doped Lithium Titanate Transport Properties and Compensation Mechanism Investigation

Led a comprehensive study of MoO₃-doped lithium titanate (LTO) to quantify how dopant level and processing conditions influence transport properties and microstructure, proposing a compensation mechanism supported by XPS alongside electronic and ionic conductivity measurements. Using XRD, Raman, SEM/TEM, EDX, TGA, and BET, I mapped structure–property relationships and benchmarked materials, additives, and solvents through analytical and electrochemical tests to assess performance potential. In parallel, I established and managed a battery research laboratory—specifying and commissioning equipment, creating fabrication and test protocols, and overseeing maintenance and troubleshooting—while coordinating multi-project teams that produced four peer-reviewed publications.

Education

Ph.D. in Materials Science and Engineering / Clemson University, Clemson, SC

(Jan 2015 – May 2020)

Dissertation: Engineered Porous Electrode for High Performance Li-ion Batteries. Among the top 10% of PhD students, graduated with GPA = 4.0. Bishop Fellowship recipient (2019-2020). Postdoctoral Fellowship recipient.

MSC. in Materials Science and Engineering / Sharif University of Technology, IR

(Sep 2010 – Jan 2013)

Dissertation: Thesis: Synthesis of CNT/CNF Nanofibers Through Electrospinning